

Rebooting McGill's Neotropical Environment Courses in Panama



After two years of pandemic lockdown, the desire for experiential learning, and for face-to-face interaction between people has never been more apparent. It is in this context that Professor Brian Leung (Bieler School of Environment and Department of Biology) and his colleagues “rebooted” their Panama graduate-level field courses: the BSE’s “Foundations of Environmental Policy” course (ENVR 610) and Biology’s “Neotropical Biology, Environment, and Conservation” course (BIOL 640), in collaboration with the Smithsonian Tropical Research Institute (STRI) in Panama City.

These two courses form the core for the Neotropical Environment Option (NEO) program at McGill, which is directed by Professor Leung. This year, they decided to use the disruption caused by COVID-19 to re-envision the courses and implement a more ambitious, integrative experience between the natural and social sciences, which has been discussed in NEO for years, and is central to the BSE’s vision. “While BSE has long recognized the importance of interdisciplinarity, implementing it in experiential learning courses requires substantial planning, coordination and investment by a dedicated teaching team,” explains Professor Leung. This year, for the first time, Instructors Holly Cronin (ENVR 610) and Catalina Gomez (BIOL 640) attempted to do so through designing an intensive month of field-based immersion to challenge students to understand and engage

meaningfully with Panama’s entangled social-ecological systems. This program was developed in close collaboration with Owen McMillan, Jarrod Scott, Benita Laird-Hopkins, Lady Mancilla and Adriana Bilgray from STRI and supported by McGill professors Catherine Potvin, Daviken Studnicki-Gizbert and Julie Major.

The NEO students traveled together across Panama to learn about ecology, conservation and policy across biomes, from terrestrial forests to marine ecosystems, and the diverse research that is done across the country. They went to Barro Colorado Island, which was formed when Lake Gatun was created from flooding due to the creation of the Panama Canal, and where tropical forest research has been conducted for 100 years. They visited the Indigenous Ipetí- and Piriati-Emberá communities, where the Bayano-McGill Reforestation Project to offset carbon generated by the university’s air travel is currently ongoing in partnership with McGill’s Office of Sustainability.

Crossing the Continental Divide, they studied a variety of agricultural systems for cultivating coffee, bananas and cacao. They explored mangroves and coral reefs and investigated issues of marine conservation, fisheries policy and waste management in archipelagos on both coasts of Panama: the Caribbean’s Bocas del Toro and the Pacific’s Coiba National Park, which has evolved from a

penal colony to a marine biodiversity sanctuary. In each site, Cronin and Gomez, in concert with local stakeholders and community members, introduced both ecological and social issues and highlighted their points of confluence. “The immersive, experiential nature of the NEO courses emphasized for the students that their work touches upon not just environmental conservation, but also real communities, people and their livelihoods,” says Leung. Consistent with BSE’s mission, this consideration across disciplines and perspectives allowed the students better position their own research and understand the context of their work and studies.

Photos by Holly Cronin, Ariadna Mora, Owen McMillan



Panama Canal seen from Barro Colorado Island



Boat drivers prepare to take students to Coiba National Park



Bayano-McGill Reforestation Project leaders in Piriati-Emberá territory



STRI Staff Scientist Joe Wright introducing ForestGEO plot on Barro Colorado Island